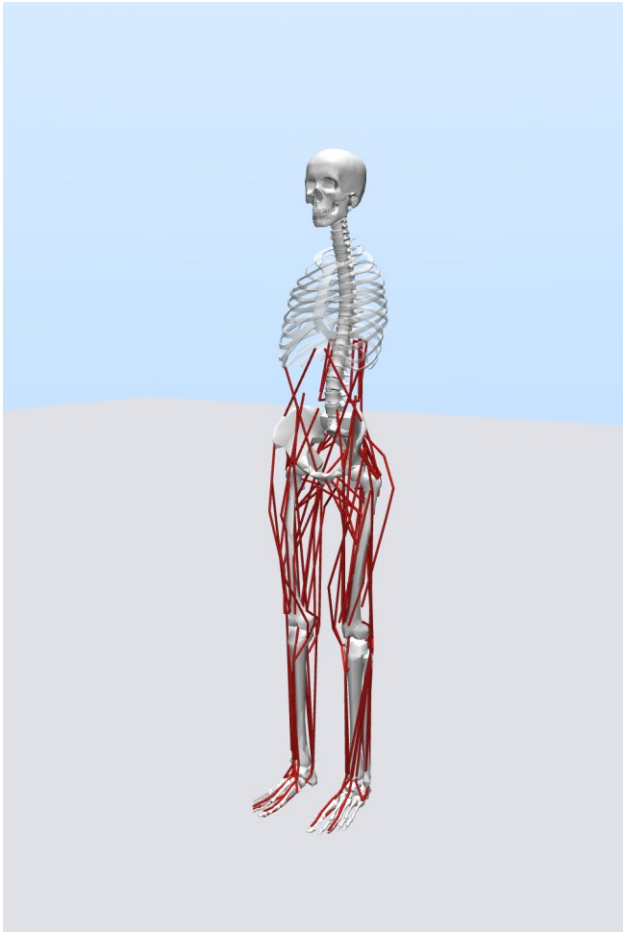
DISSERTATION DEFENSE

Movement and Control of Biomimetic Humanoid Robots

Ben P. Bolen

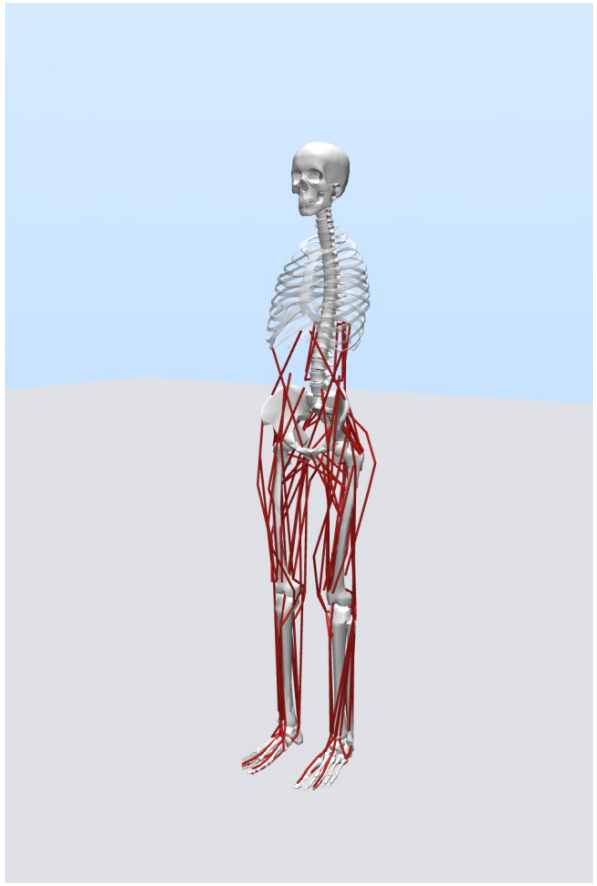
Mechanical Engineering PhD defense
Advisor: M. Anthony Hunt, Ph.D.
November 6, 2026



Four gaps connect actuator mechanics to neural-control experiments

Knowledge gap	Contribution tested here
Force across actuator lengths	Resting-length-aware force model
Torque through compliant routes	Identified corrections and redesigned-route tests
Scattered sensory evidence	Linked, searchable SADb resource
Untested model connections	Interfaces with matched circuit checks

This dissertation adds models and tests to an existing robot program

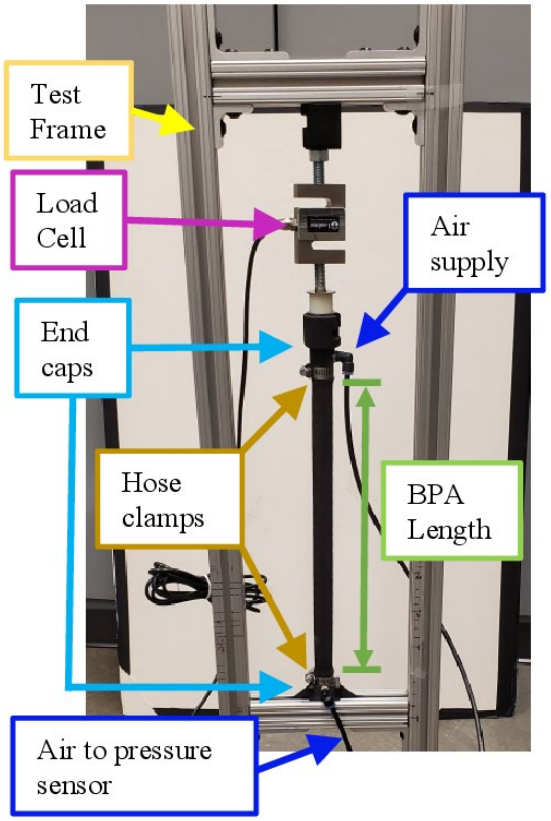


Existing anatomy and mechanisms



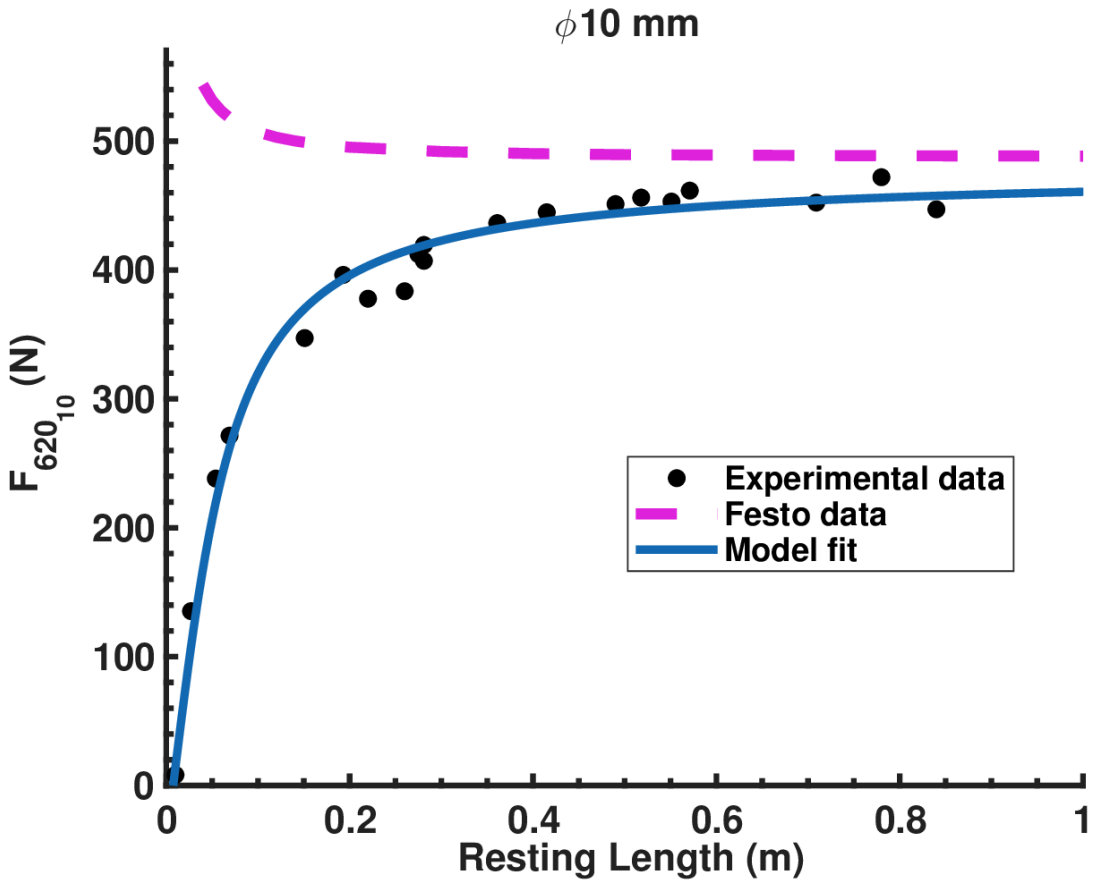
New route modeling and tests

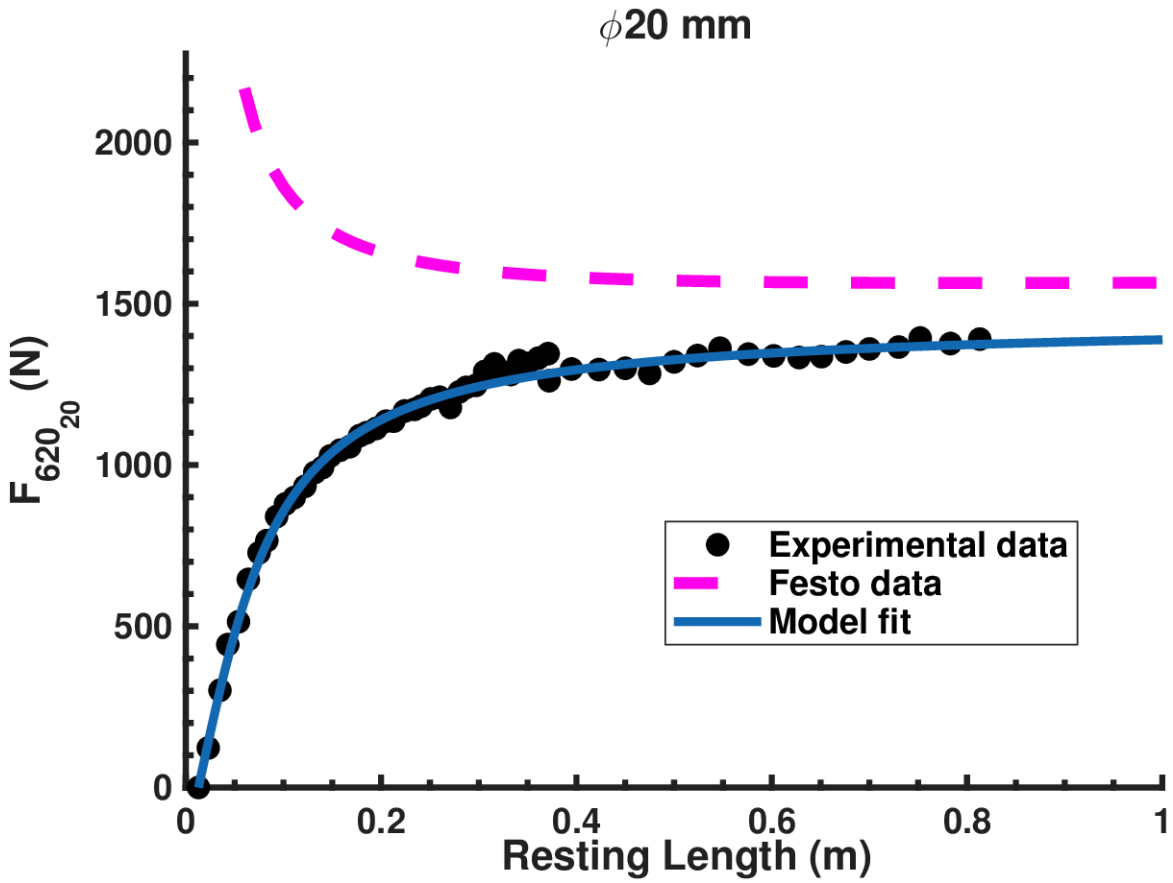
Isometric tests measured force across length, pressure, and contraction



Braided pneumatic actuators (BPAs): 537 observations

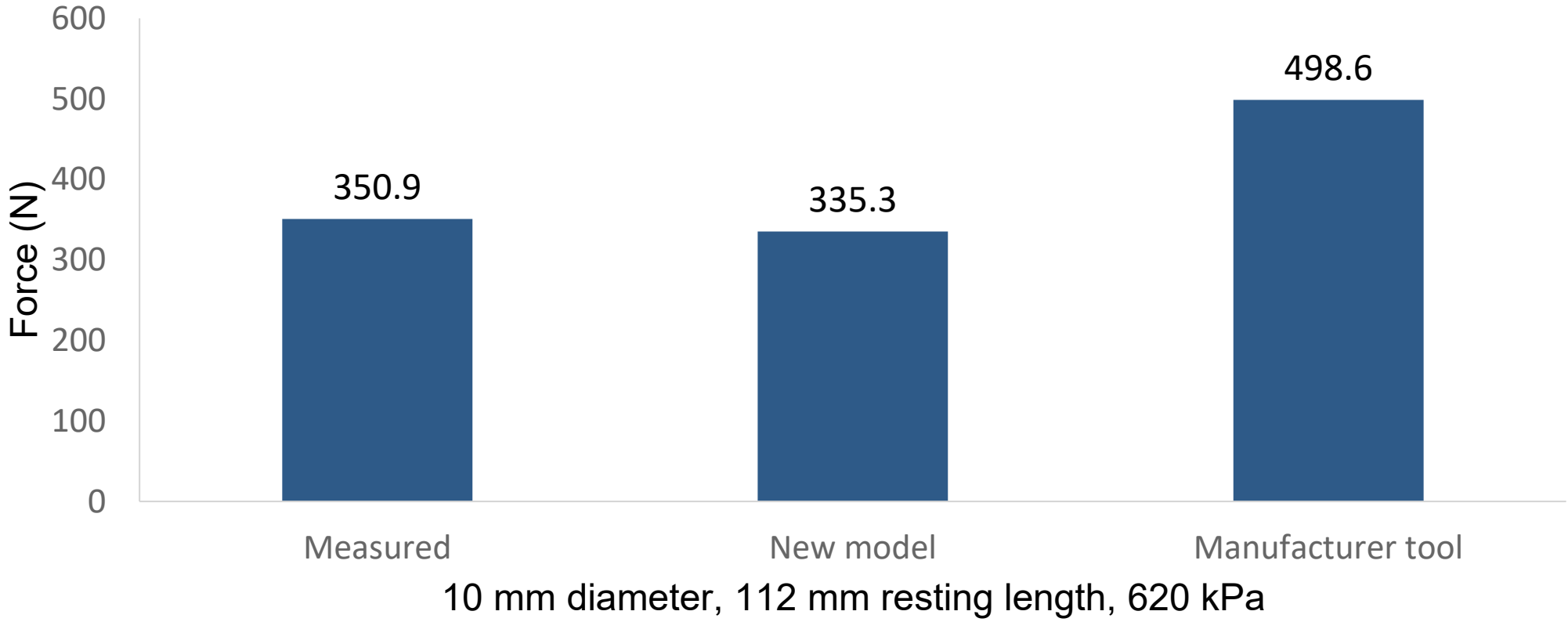
Maximum isometric force decreases as resting length becomes shorter





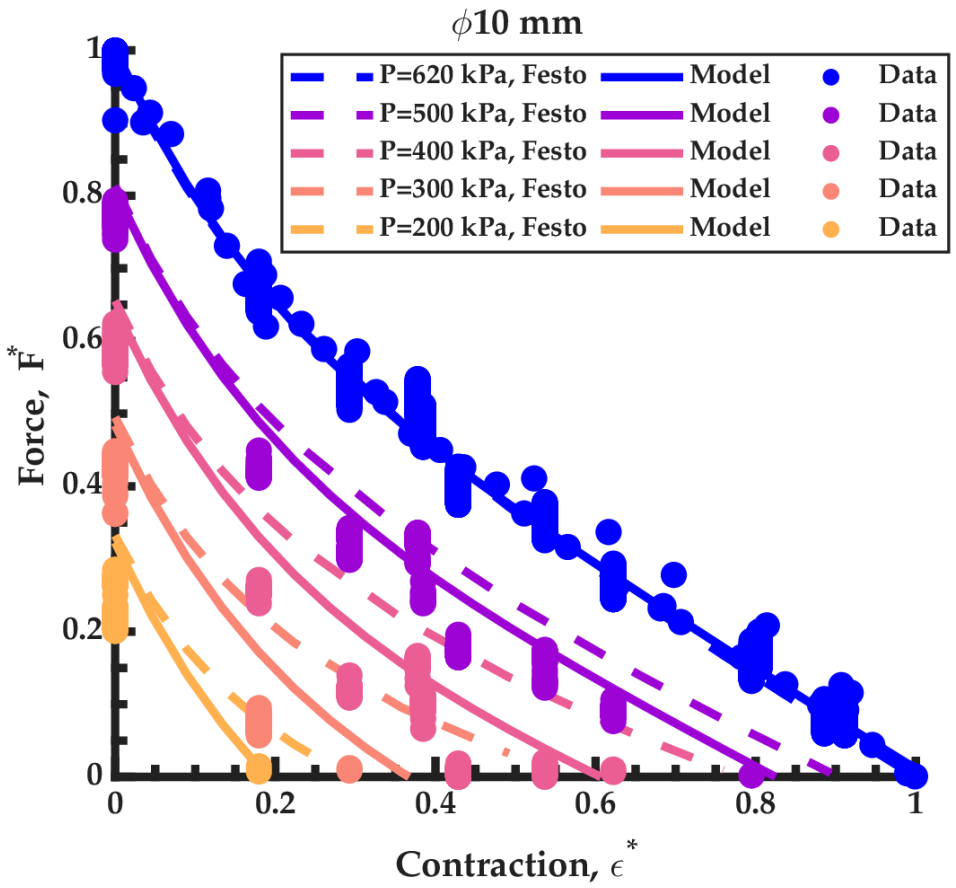
Bolen et al. (2026), Actuators · Revision C, Fig. Fmax_fit

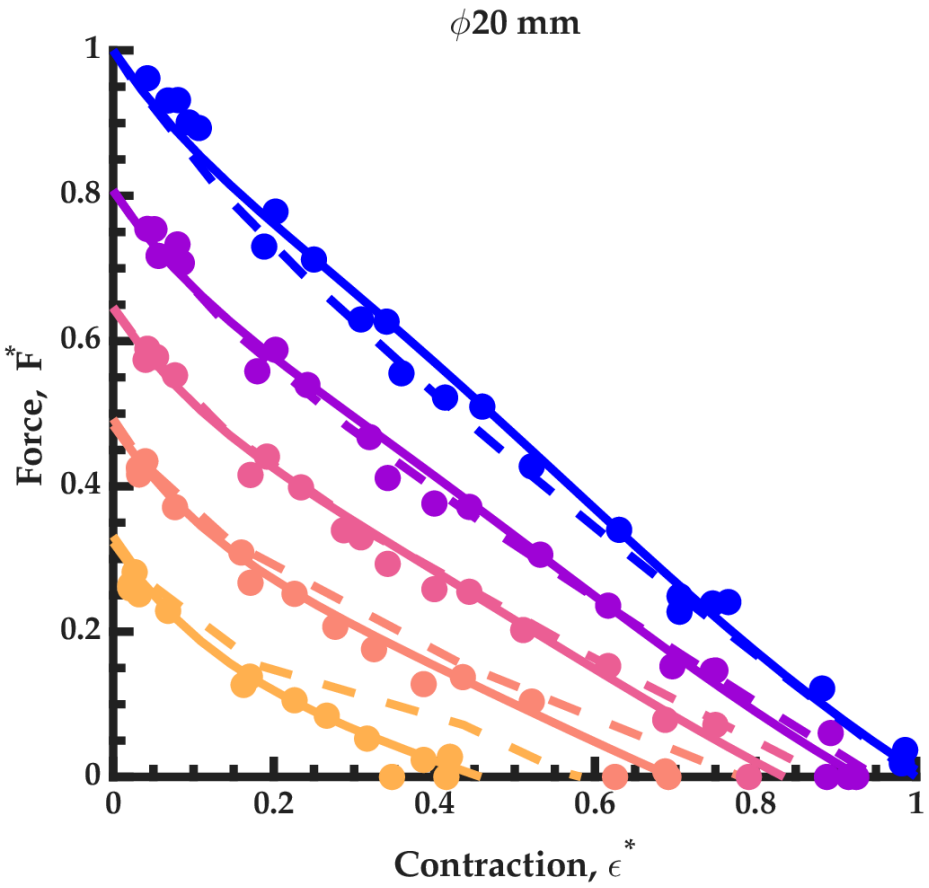
At 112 mm, the new model reduces force error from 42.1% to 4.5%



350.9 N measured

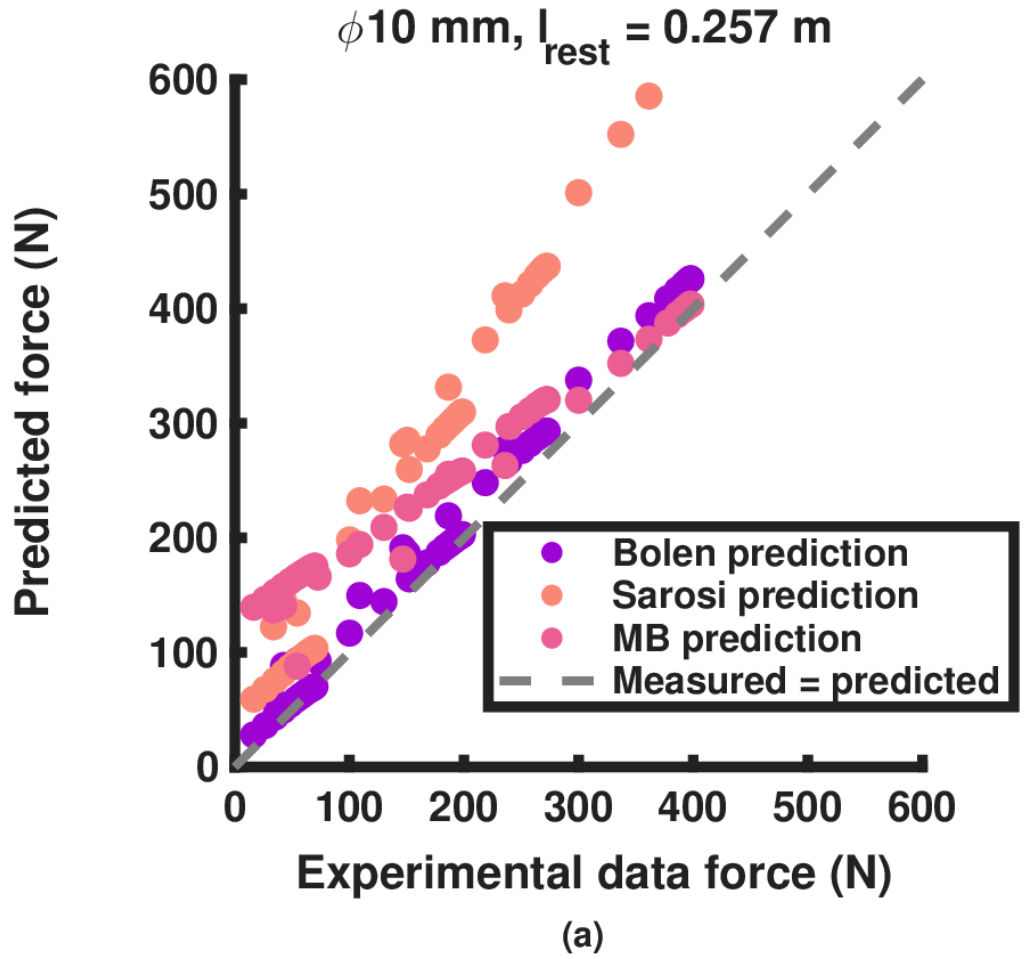
Normalized force curves describe both tested actuator diameters

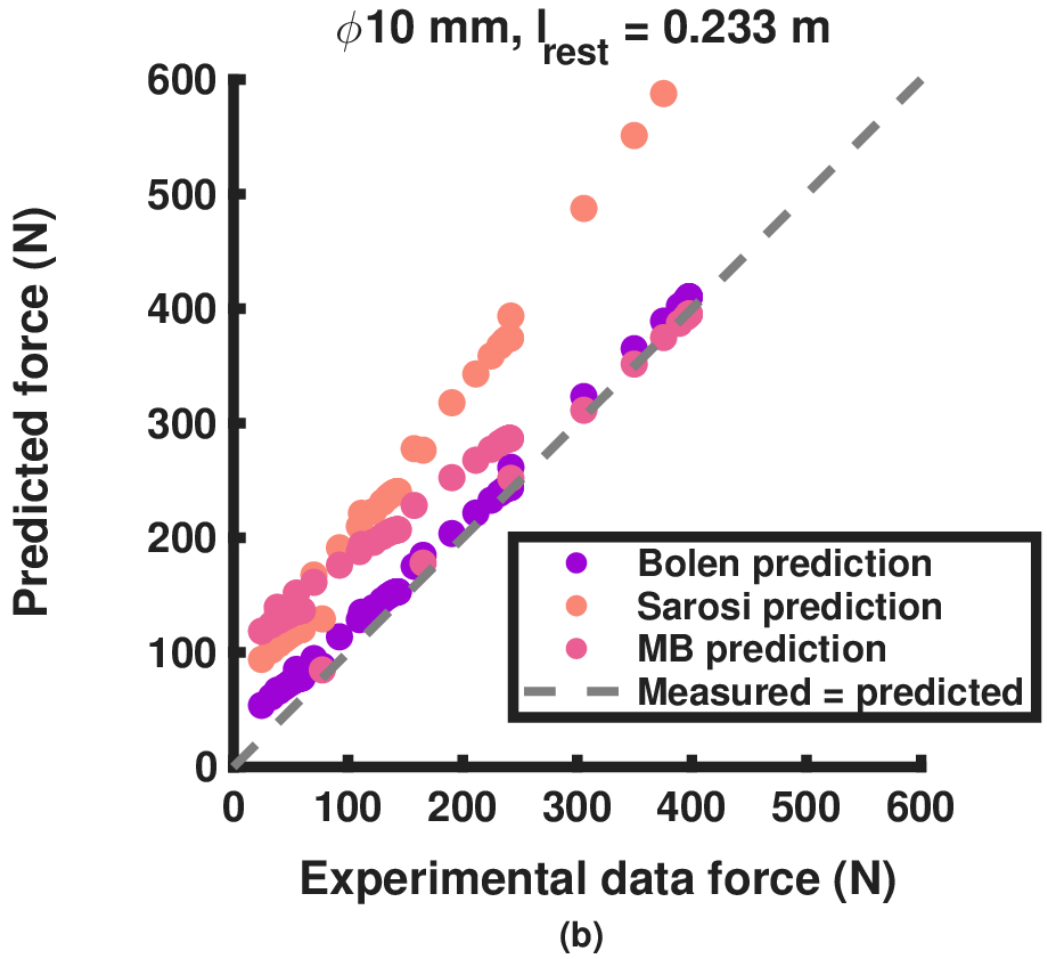




Bolen et al. (2026), Actuators · Revision C, Fig. GenForceFit

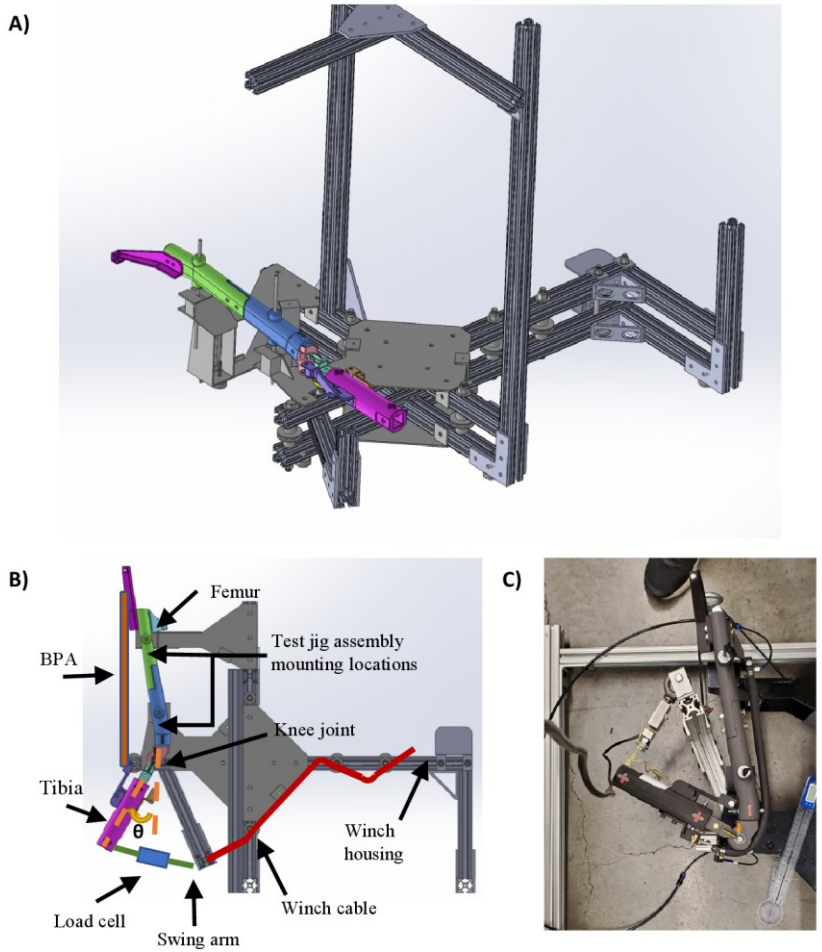
The new model has the lowest error in the reported model comparisons





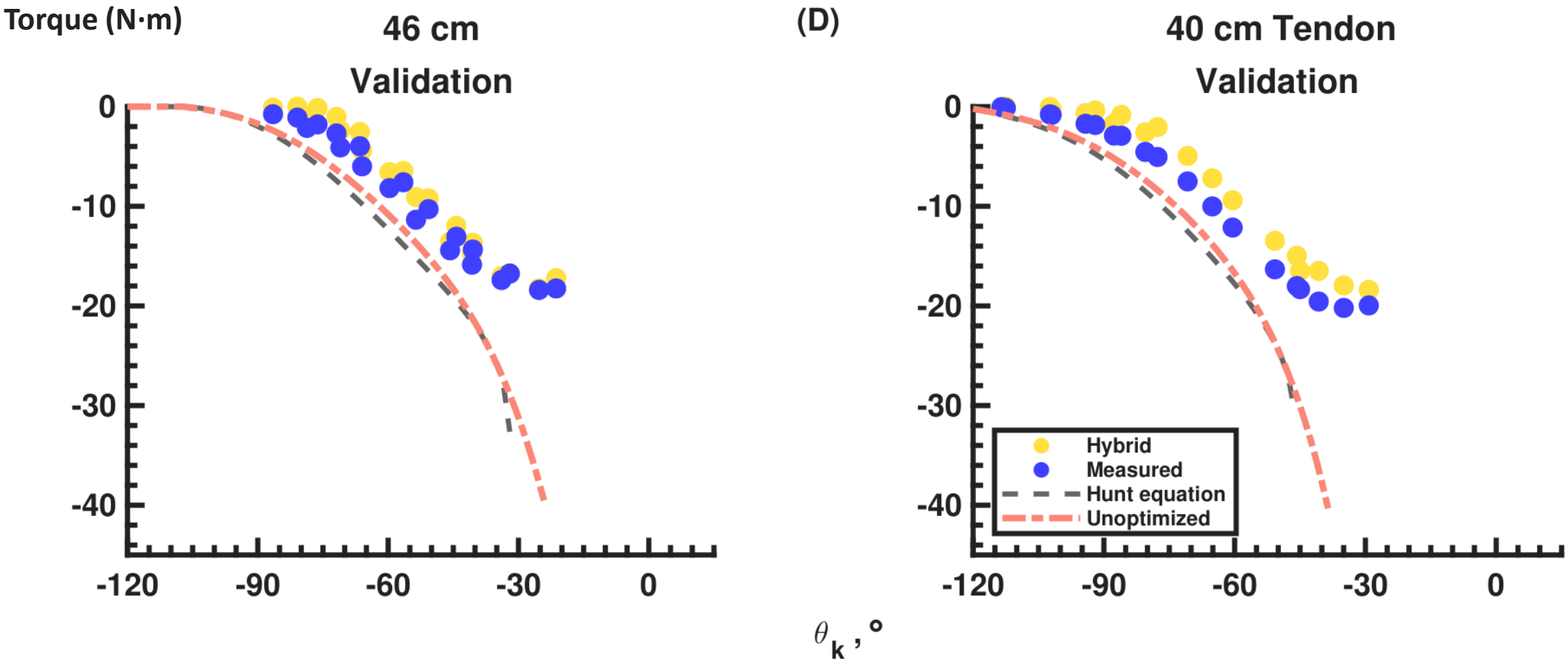
Bolen et al. (2026), Actuators · Revision C, Fig. Comparison and model-comparison table

Pinned and biomimetic knees separate identification from transfer tests



Revision C · Ch. 4, test stands; four-bar knee credited to Steele et al.

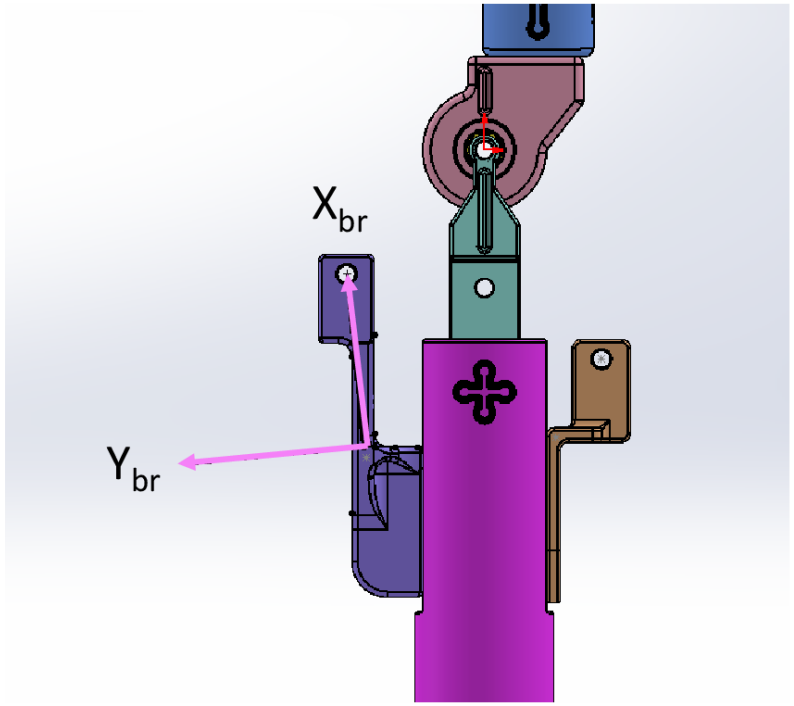
Measured actuator length exposes errors in the rigid-route prediction



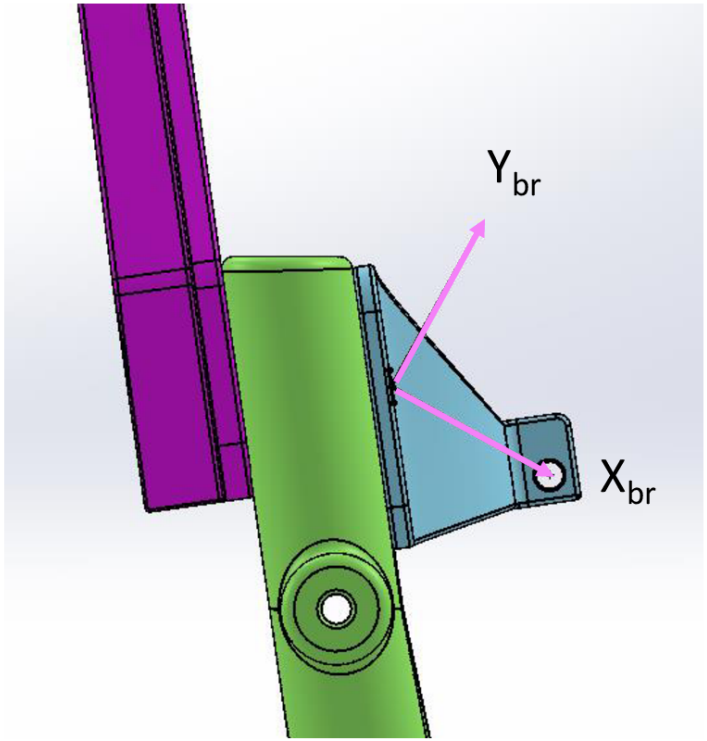
Revision C · Ch. 4, Fig. FlxPin_preopt; hybrid-method description

The corrected model accounts for length offset, compliance, and wrapping

(A)



(B)



Effective system stiffness includes the fixtures and transmission

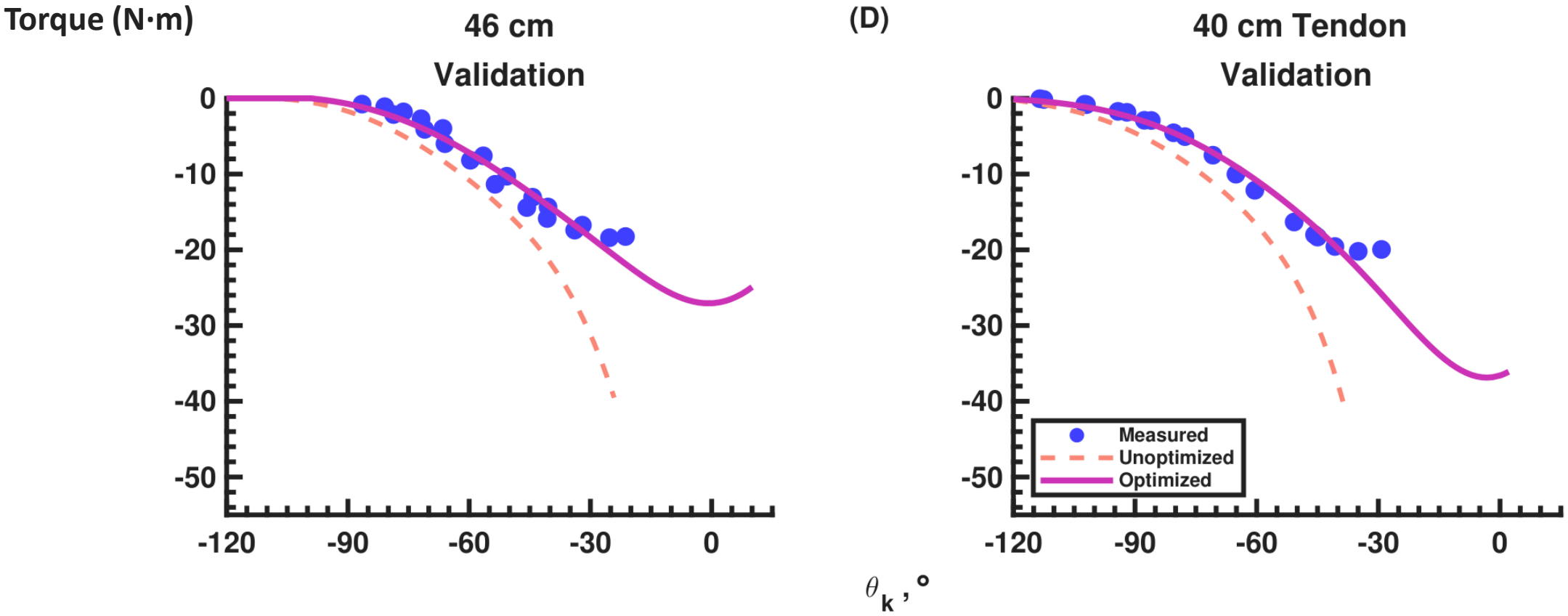
Identification reserves tests before assessing corrected predictions

Adopted fold	Training	Held out	Not used
Pinned flexor	2 tests	2 tests	1 test
Pinned extensor	3 tests	3 tests	3 tests

Identify corrections: gamultiobj

Redesign routes: surrogateopt, then patternsearch

The corrected flexor model reproduces the held-out torque trend



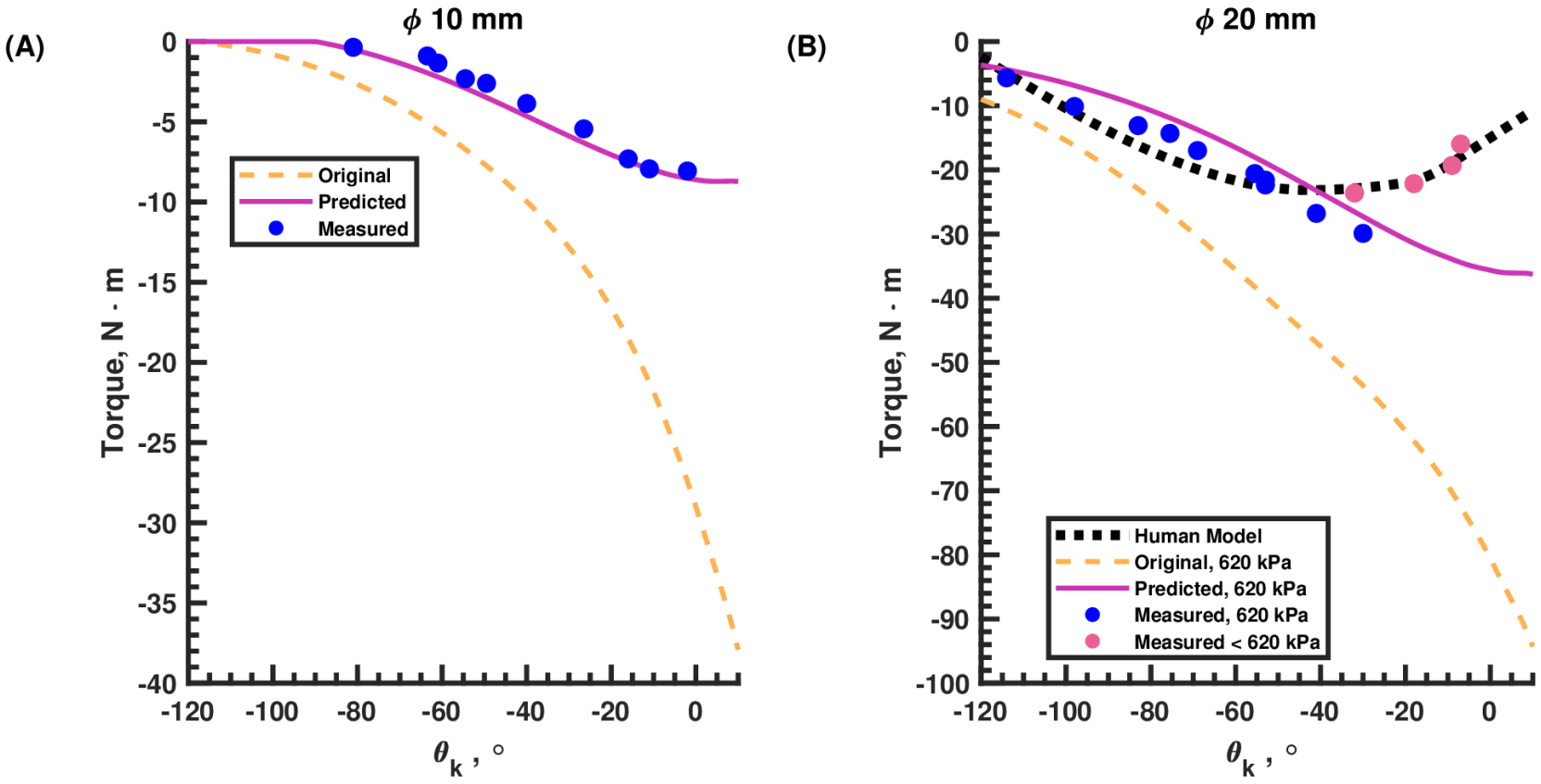
Held-out root-mean-square error: 7.588 to 1.490 N·m

Revision C · Ch. 4, Fig. FlxPin_optimized; Table FlxPinGof

Corrections reduce held-out error in both pinned-knee configurations

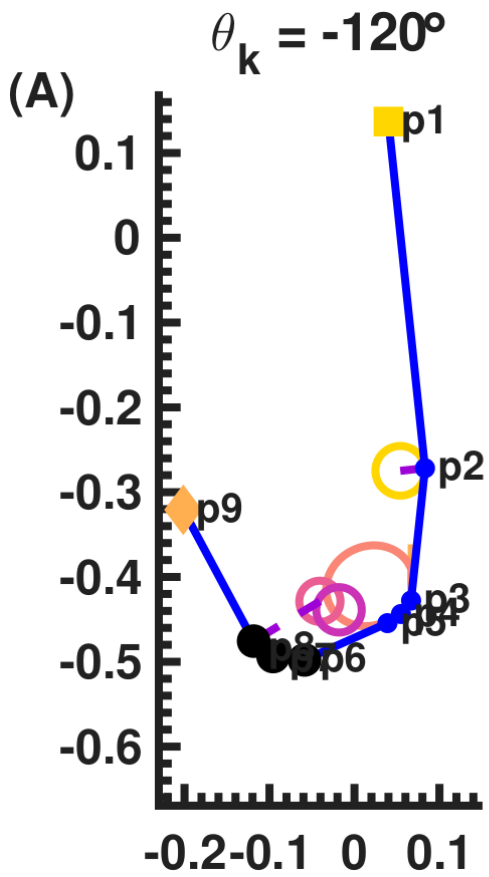
Root-mean-square error (N·m)	No corrections	Corrected
Flexor, 2 validation tests	7.588	1.490
Extensor, 3 validation tests	2.554	1.098

The earlier single-actuator flexor falls short of its human target



Earlier geometry: one 20 mm actuator

The optimized cam route became a two-actuator test assembly



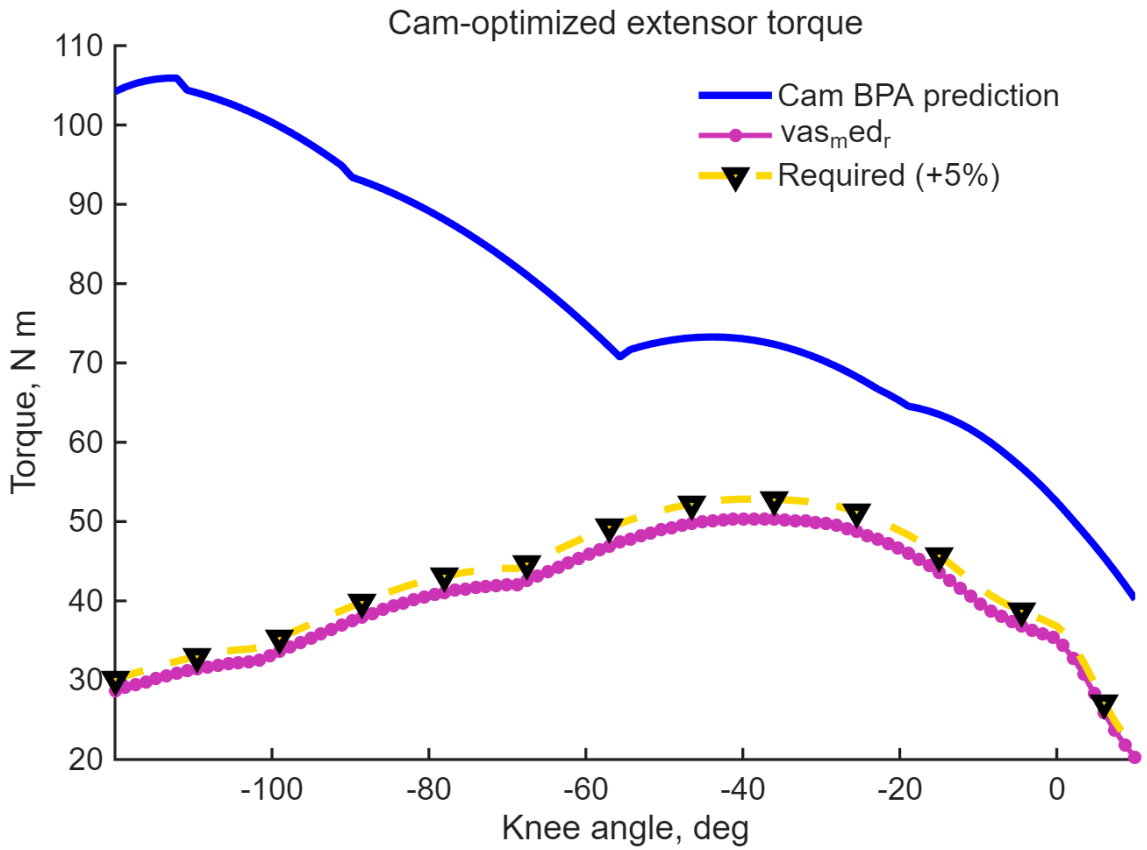
Route at −120°; coordinates in m



Assembled cam-extensor hardware

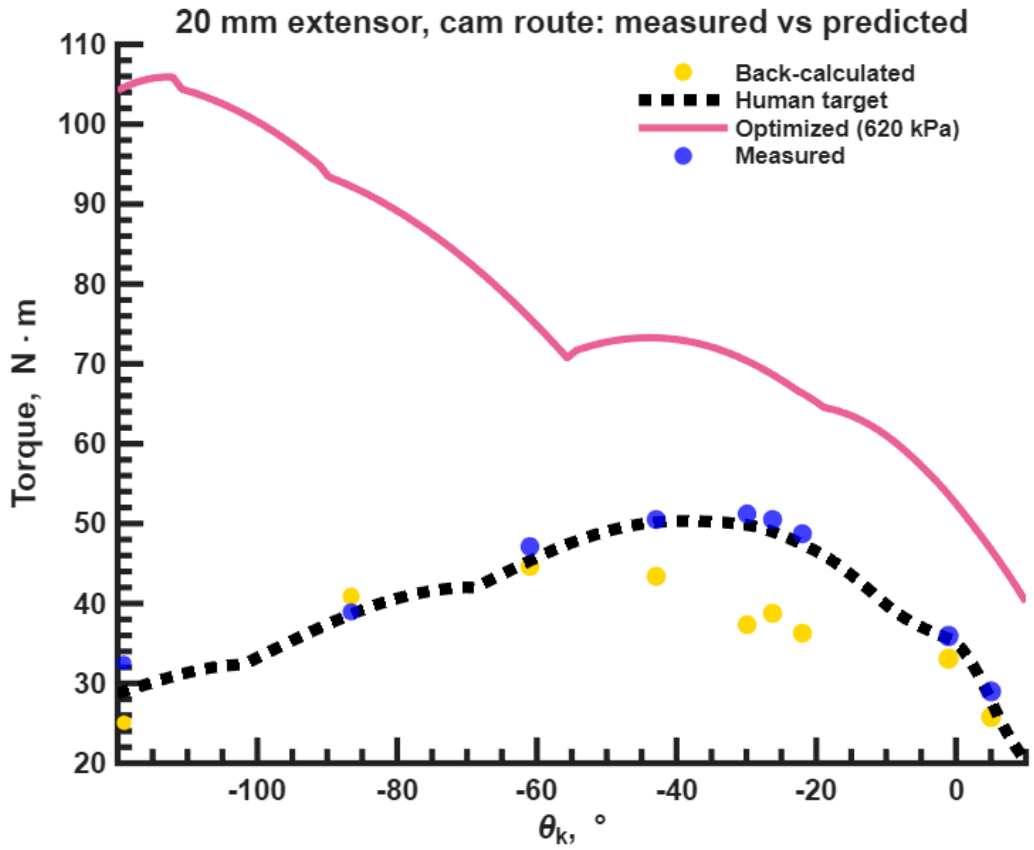
Revision C · Ch. 4, Cam route, panel A; assembled hardware photograph

The cam model predicts at least 30.83% margin over the required envelope



Prediction at 620 kPa; required envelope includes 5% overshoot

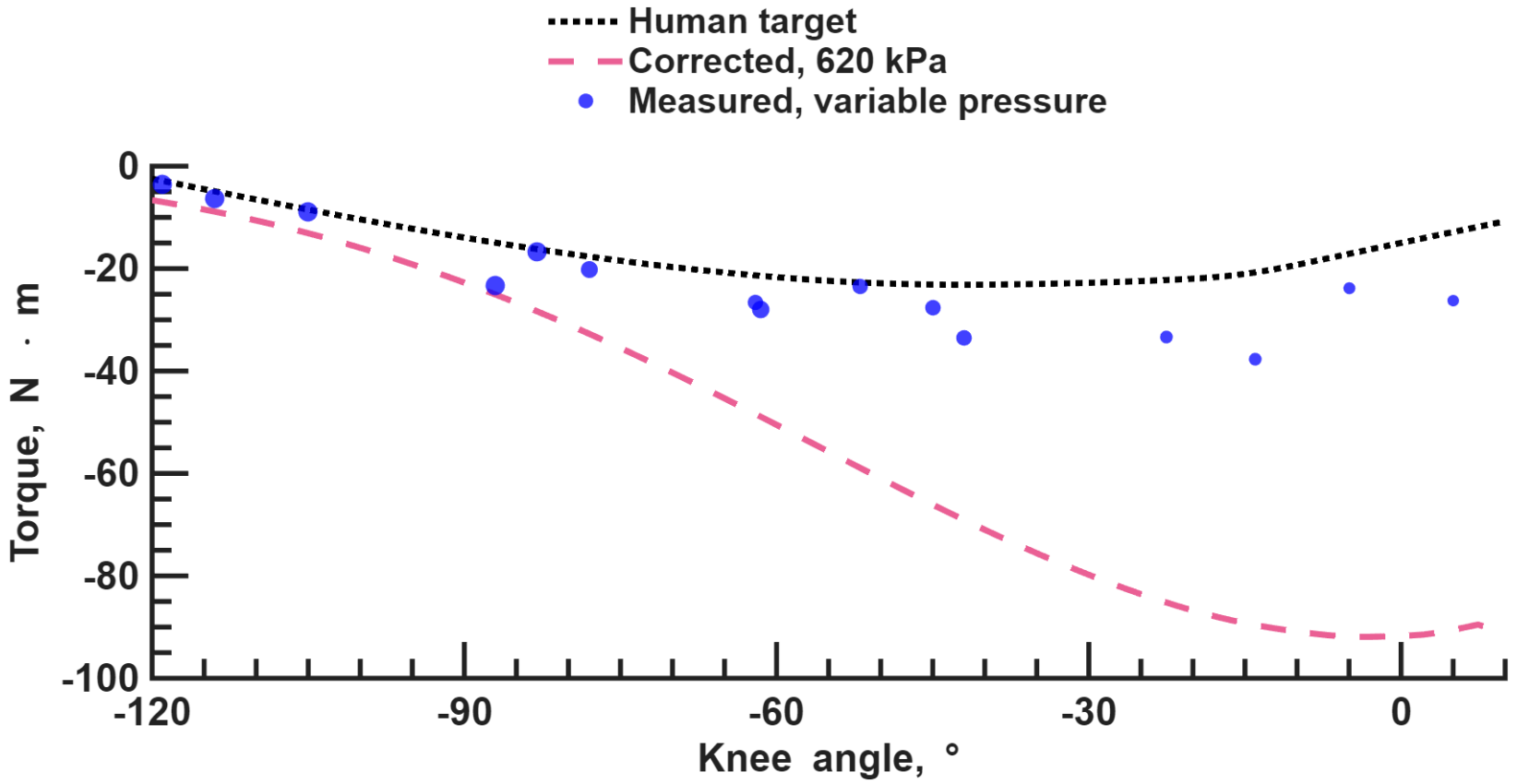
Nine cam-extensor holds test torque across −119° to +5°



Pressure varied between holds

Revision C · Ch. 4, Fig. extensor_physical_final; MATLAB Ext20mm_compare_20261022.png

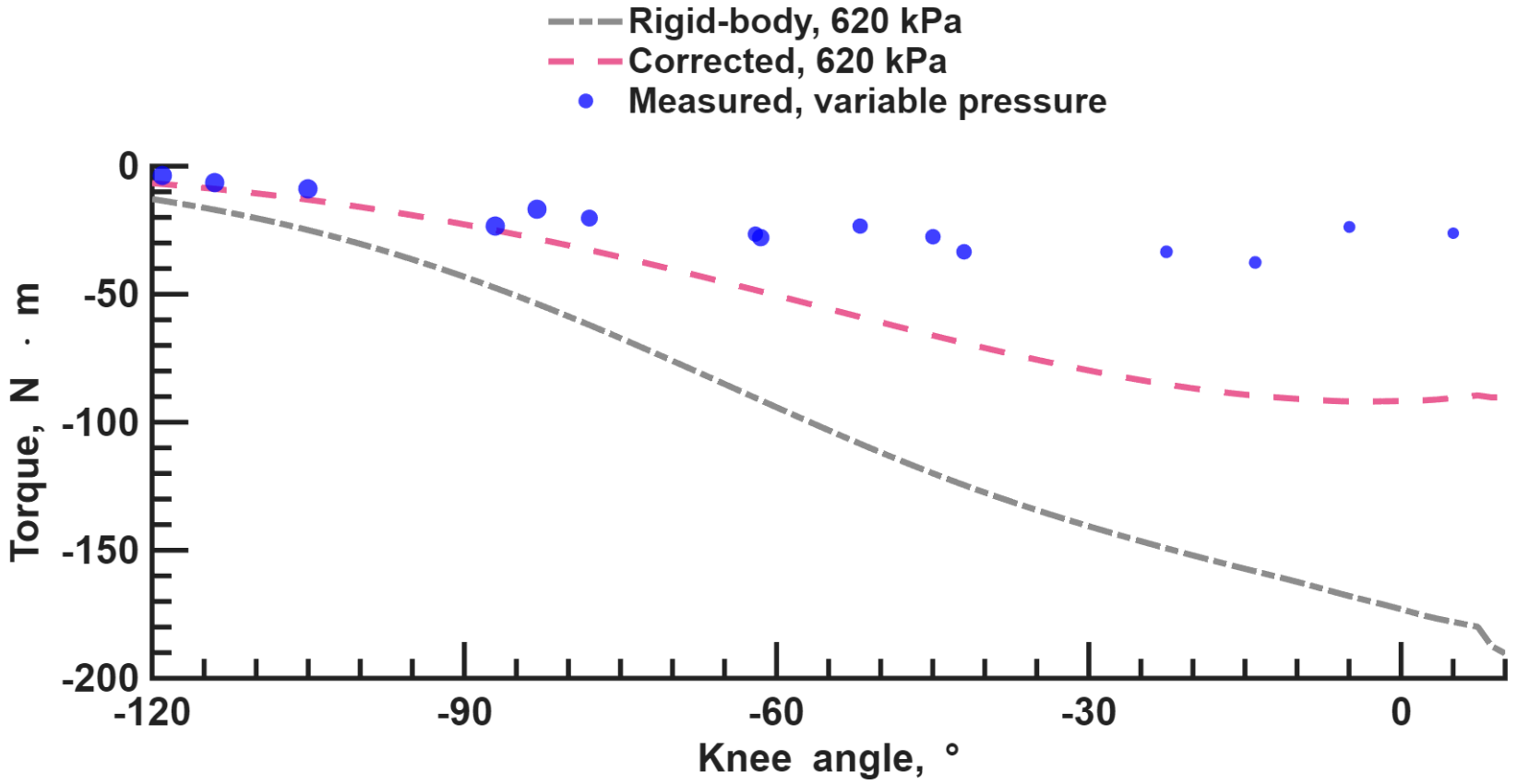
All 15 flexor holds meet or exceed the human muscle target



15 holds at 226.5–638 kPa; margins +0.42 to +17.0 N·m

Knee-torque study · Fig. Flx20mm; final MATLAB reduction

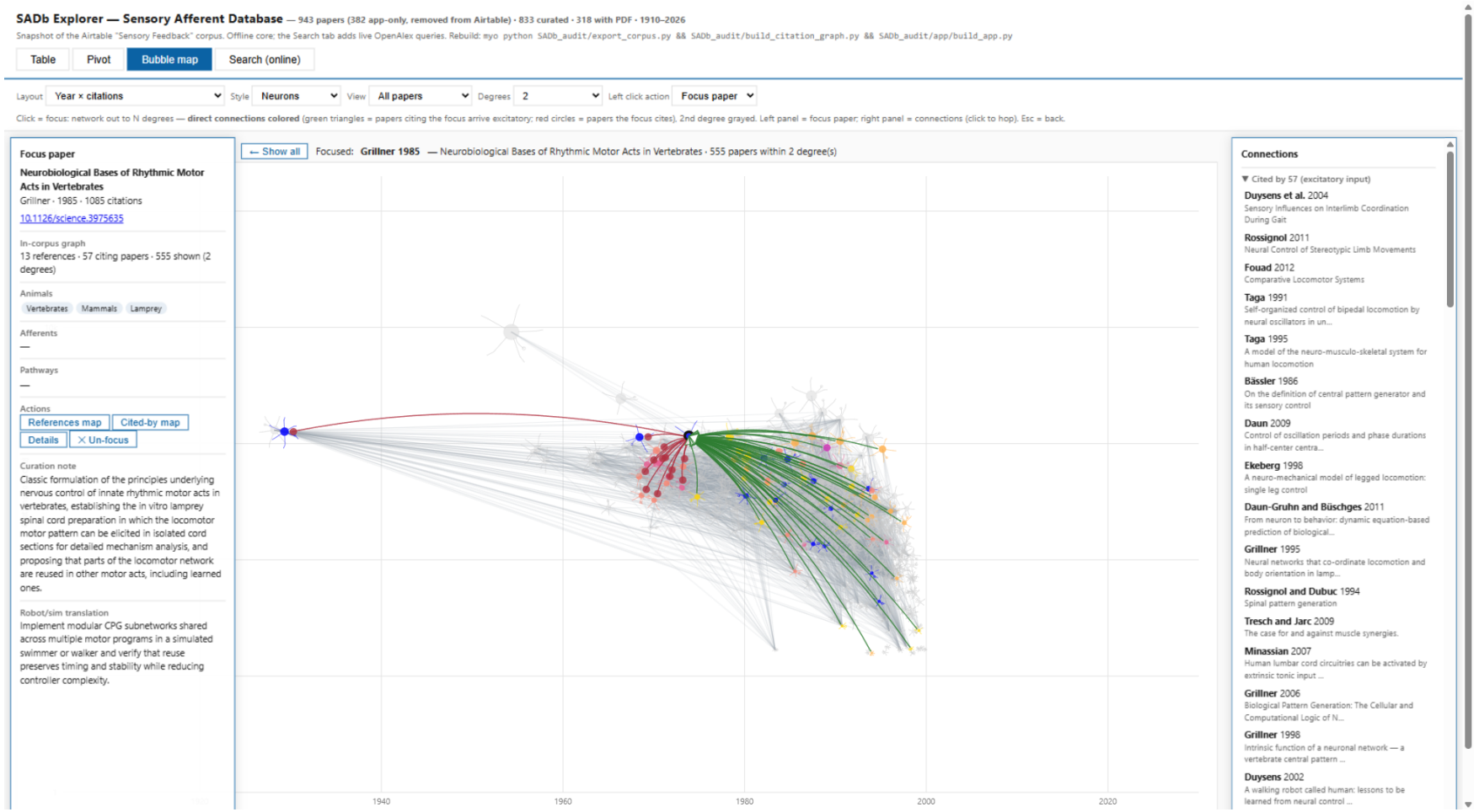
The corrected 620 kPa prediction still exceeds many measured flexor torques



Fixed-pressure curves; variable-pressure measurements

Knee-torque paper · Fig. Flx20mm, panel B; unchanged MATLAB calculation

SADb links sensory-feedback literature to inspectable source records



Revision C · Ch. 5, The Resource; SADb application snapshot

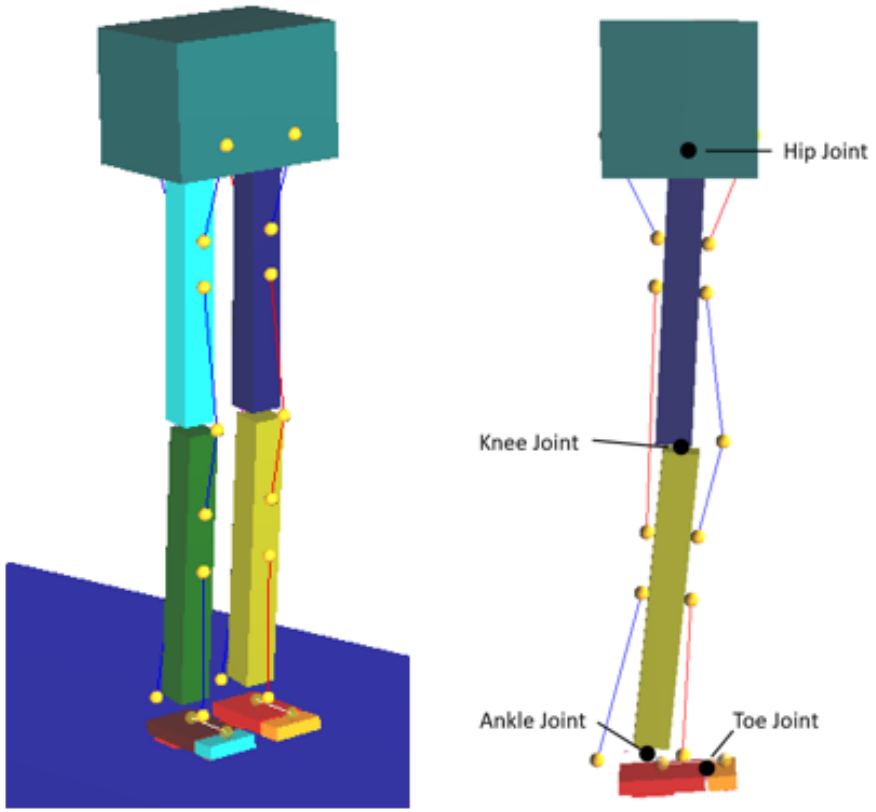
SADb exposes classification coverage without supplying controller gains

Rows: Afferent types | Columns: Animal | Sort rows: by count | Sort cols: by count | Count = pap

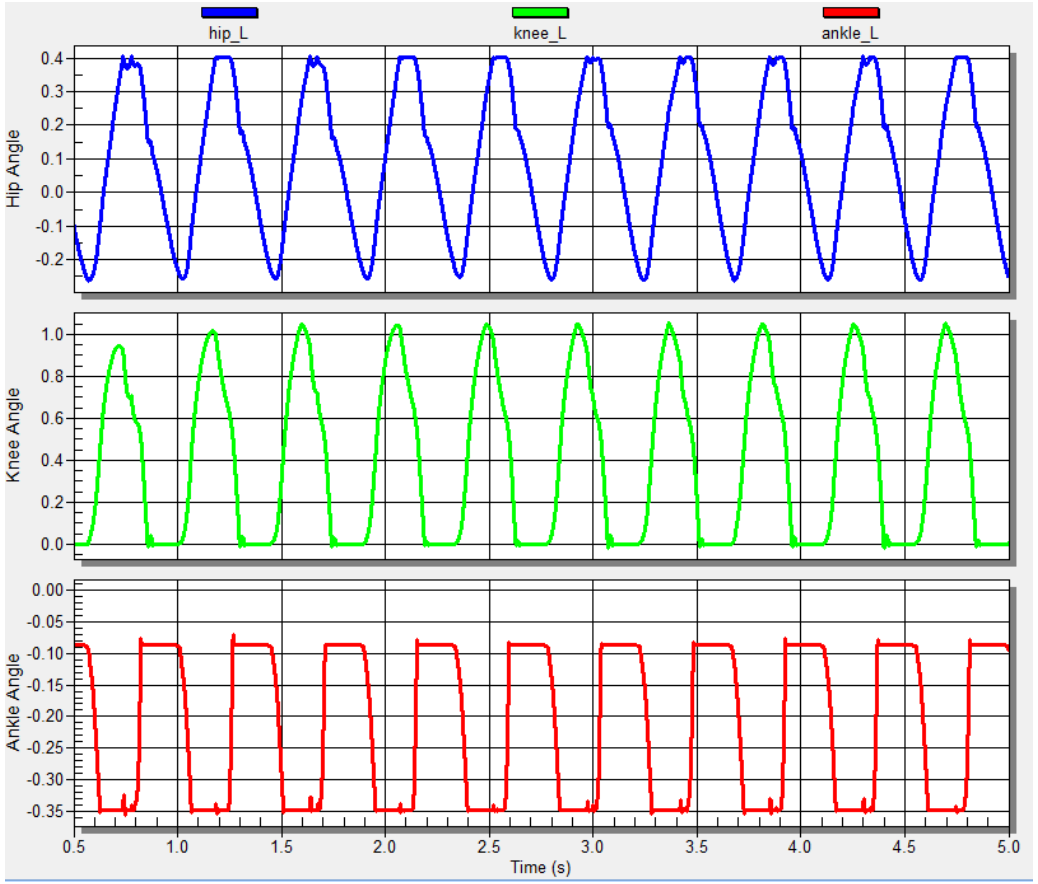
Afferent types ↓ / Animal →	(untagged)	Cat	Human	Mice	Mammals	Insects	Vertebrates	Rat
(untagged)	327	162	202	64	51	40	31	29
Cutaneous	5	28	6	1	2	·	1	·
Ia	4	29	5	2	·	·	1	1
Ib	5	21	7	·	3	·	·	1
II	2	16	6	·	1	·	1	1
Mechanosensory	·	·	·	·	·	2	·	·
Flexor reflex afferents	1	4	·	·	·	·	·	·
III/IV	·	1	·	·	·	·	1	·
Nociceptive	·	1	·	·	·	·	·	·

Tagged papers retain their experimental context

The initial AnimatLab controller used one rhythm generator for suspended stepping



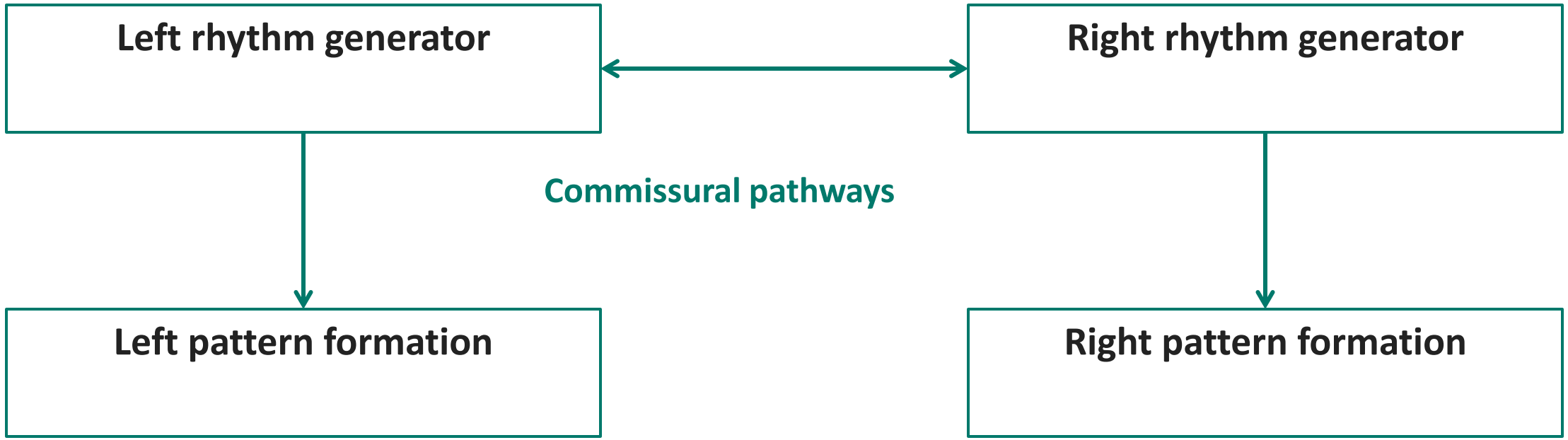
Historical mechanical model



Original left-joint traces

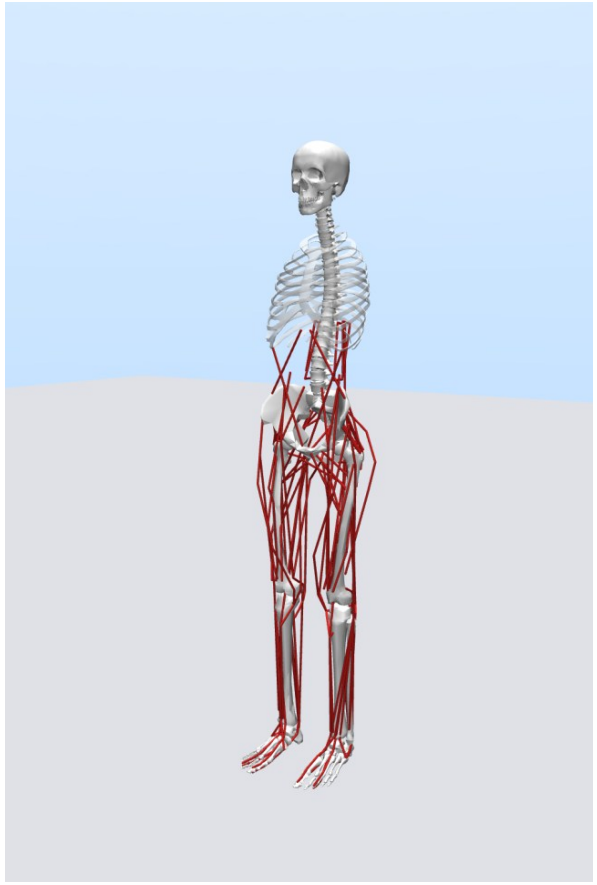
Revision C · Ch. 6, The AnimatLab Bipedal Walker; Li (2017), Deng (2019), Bolen (2021)

I later interconnected separate left and right rhythm generators

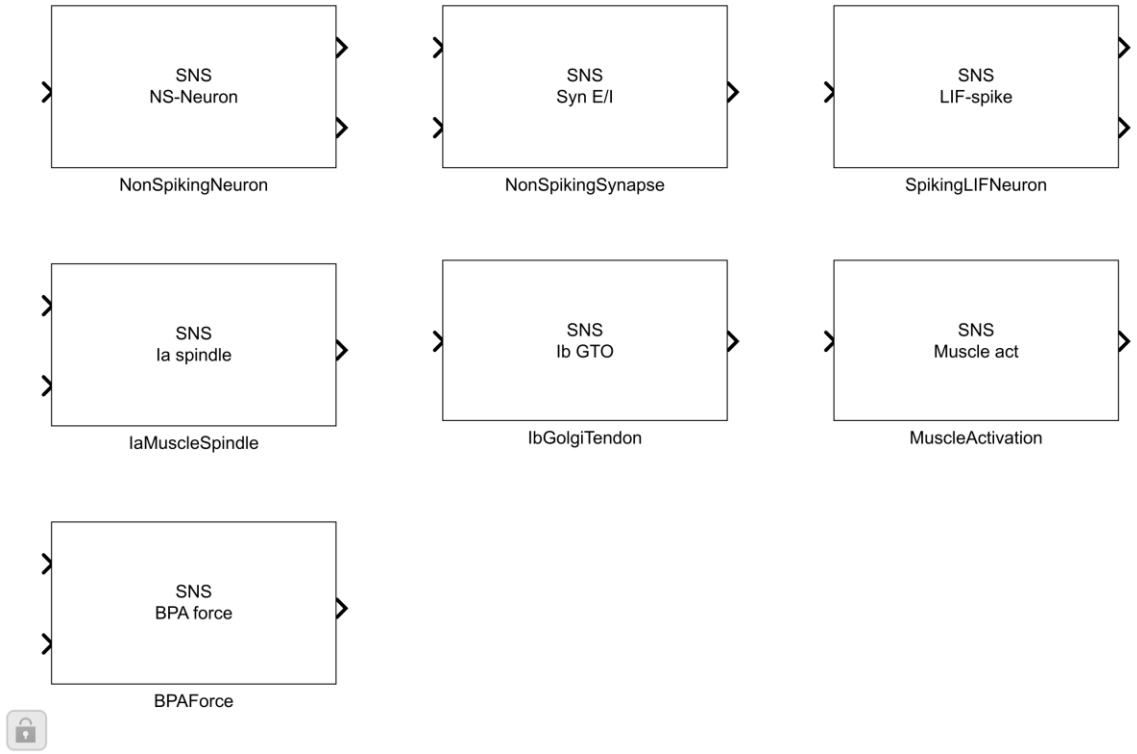


Conceptual architecture; walking remains unverified

My interfaces connect existing mechanics and neural formulations

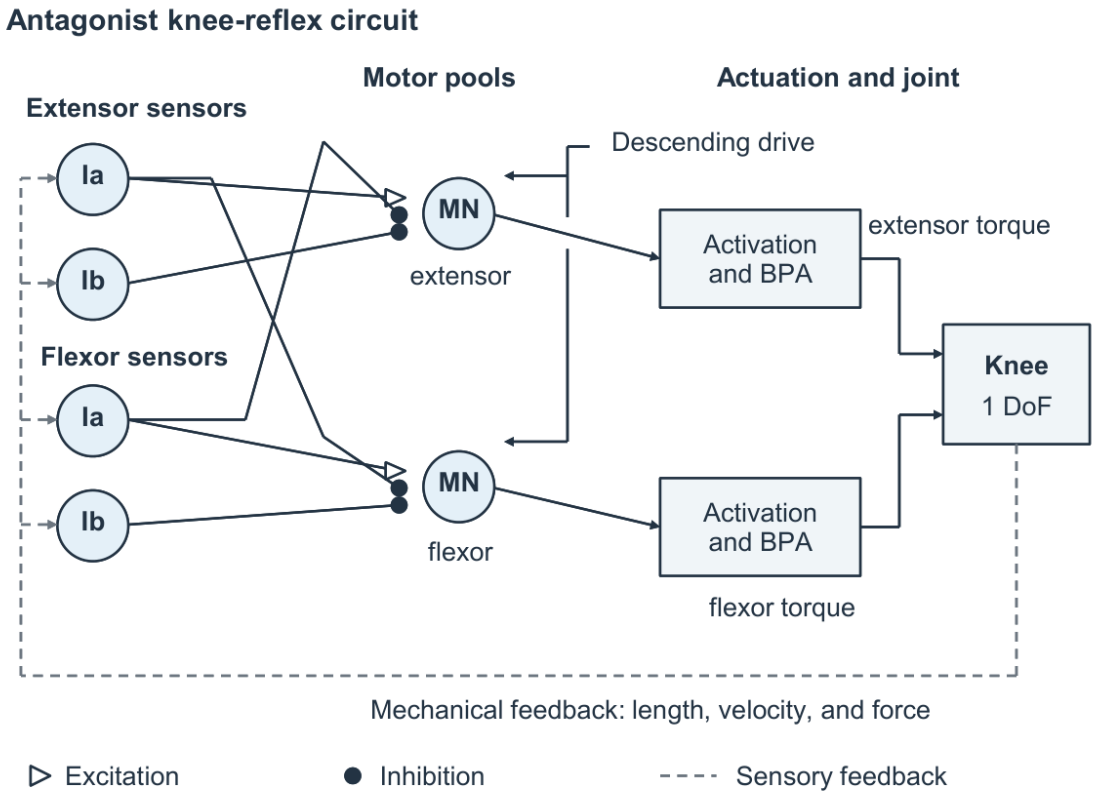


Repaired and audited model coupling



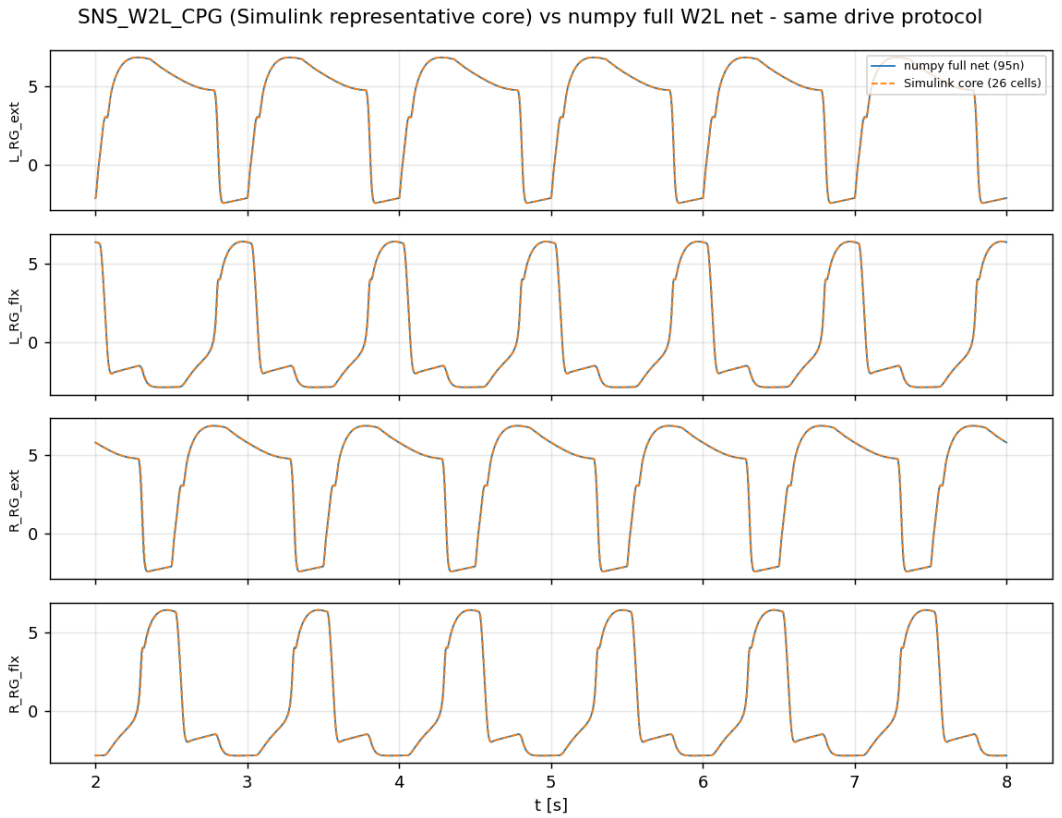
Editable implementations of existing equations

The graphical library implements defined Ia and Ib reflex pathways



Spindle (Ia) and tendon-organ (Ib) inputs in a provisional model

Matched rhythm-core implementations reproduce the same driven bursts



1.000 s periods; trace correlation ≥ 0.99996

Revision C · Ch. 6, Fig. w2l_crossengine; matched NumPy and Simulink protocol

The bridge exchanges signals while full-network dynamics still disagree

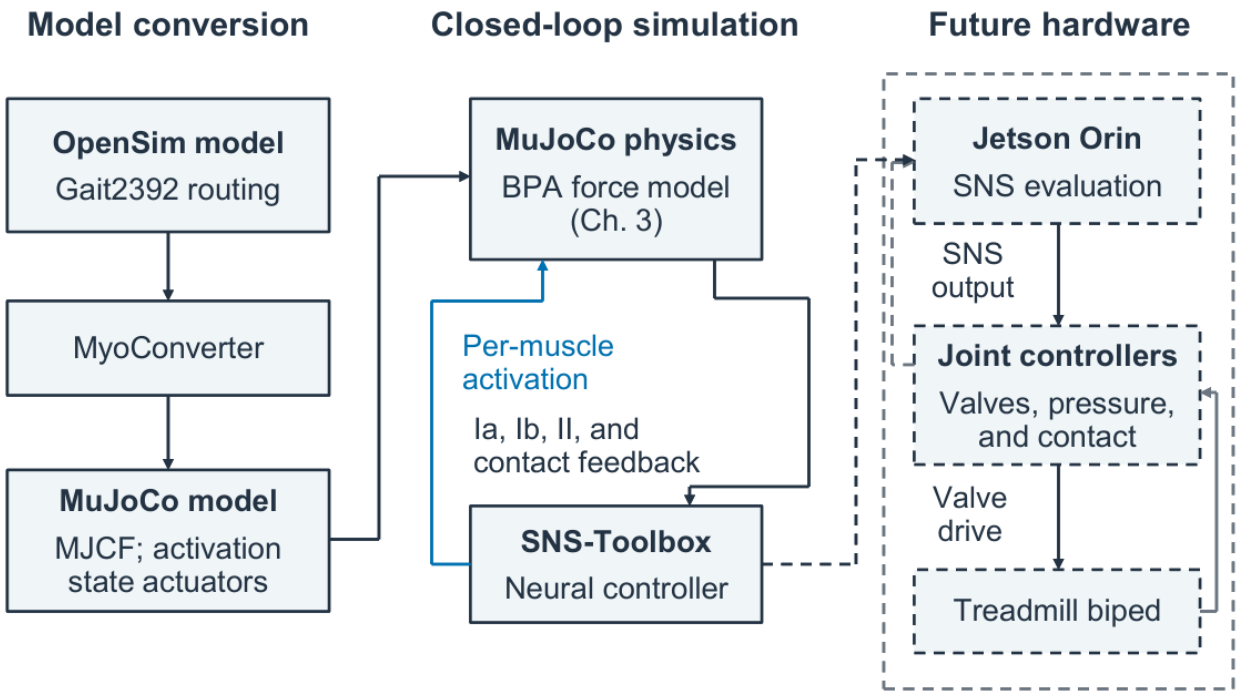
Verification level	Observed result
Sensor and command exchange	Bit-exact scored readback
92-actuator closed-loop run	Signals exchanged for 5 seconds
Full network under tonic drive	Simulink settles; NumPy oscillates

Interface correctness and controller equivalence need separate tests

The evidence supports static torque capacity and specific simulation interfaces

Contribution	Evidence	Boundary
Actuator force model	Measured force and held-out fits	Static, tested actuator range
Joint model and routes	Held-out error and physical holds	Variable pressure; residual mismatch
SADb	Inspectable linked records	Uneven source coverage
Neural interfaces	Matched tests and signal readback	Full-network discrepancy; no free walking

Matched-pressure tests and controller reconciliation are the next checks



Computer-in-the-loop: test embedded control against the simulated body before transfer to the physical robot.

Close the measured gaps before transferring control to hardware

Questions

Thank you

Advisor: M. Anthony Hunt
Committee, AARL, and research collaborators